

Induction in action

Induction is the production of an electrical current by a changing magnetic field. Many devices, from microphones (see below) to microcomputer parts, rely on it. Computer hard disks, for instance, store data magnetically on a stack of disklike platters. The surface of each platter contains billions of individual areas, each of which can be magnetized. The patterns of magnetism store data as binary digits (1s for magnetized areas and 0s for demagnetized areas) and are produced by induction caused by a tiny moving electromagnet.

▷ Hard disk

A hard disk drive consists of a set of disks called platters. Data is written, read, or deleted by an electromagnet on a moving arm.

arm is very light so that it can flick quickly to the correct position

read-write head at the tip of the arm contains a tiny electromagnet

magnet moves the arm from side to side

disk spins at a high speed, and the head reads and writes as the surface passes beneath

a metal mesh protects the delicate diaphragm. Outdoors, special shields are used to reduce the noise of the wind blowing across it,

coil's movement within the magnetic field generates a current in the coil

magnet

sound waves make the diaphragm vibrate

diaphragm's movements move coil up and down

wires transfer current's signals to an amplifier, recorder, or loudspeaker

◁ Take a look inside

In a microphone, sound waves vibrate a diaphragm, which is attached to a wire coil. As the coil moves, it induces currents, which form a changing electrical signal.

◁ Microphone

A microphone is carefully designed so that it mimics the sounds it receives, and does not overemphasize particular frequencies.

Electromagnets are used to **lift heavy loads** of steel. The most powerful can lift single loads **weighing more than 250 tons**.

REAL WORLD

Induction cooking

The electromagnet in an induction stovetop generates a magnetic field. Some of its energy transfers to the metal pan via the process of induction, as circulating electric currents. The electrical resistance of the metal means that some of this electrical energy is converted to heat, warming the pan but not the surface.

